

Build a Graph RAG Pipeline with Neo4j in Colab

From Installation to Intelligent Q&A

Install • Ingest • Build Knowledge Graph • Index • Ask Questions • Visualize

Java 17 | Neo4j + APOC | PDF Ingestion | Azure OpenAI | GraphRAG | Insights

1 Install Java 17 (Required for Neo4j)

```
!apt-get install openjdk-17-jre-headless -qq
!update-alternatives --set java \
   /usr/lib/jvm/java-17-openjdk-amd64/bin/java
!java -version
```



✓ Neo4j 5.21 requires Java 17

2 Install Neo4j + APOC & Start Server

1. Download Neo4j
2. Extract & Rename
3. Add APOC Plugin
4. Enable APOC
5. Disable Auth (Dev)
6. Start Server

```
!wget -q https://neo4j.com/artifact.php?-5.21.0-unix.tar.gz -O neo4j.tar.gz
!tar -xf neo4j.tar.gz
!mv neo4j-community-5.21.0 nj
!wget -q https://github.com/neo4j/apoc//apoc-5.21.0-core.jar \
     -O nj/plugins/apoc-5.21.0-core.jar
!echo "dbms.security.procedures.unrestricted=apoc.*" >> nj/conf/neo4j.conf
!echo "dbms.security.procedures.allowlist=apoc.*"  >> nj/conf/neo4j.conf
!sed -i '/#dbms.security.auth_enabled/s/^#//g' nj/conf/neo4j.conf
!nj/bin/neo4j stop
!nj/bin/neo4j start
```

✓ Neo4j Community 5.21.0 is running on port 7687

3 Connect & Configure LLM + Embeddings

Driver

neo4j://localhost:7687
user: neo4j
pass: password

Azure OpenAI (LLM + Embeddings)

- GPT-4o (LLM)
- text-embedding-3-small
- Stored securely in Colab Secrets (userdata)

```
from neo4j import GraphDatabase
from neo4j_graphrag.llm import AzureOpenAILLM
from neo4j_graphrag.embeddings import AzureOpenAIEmbeddings

driver = GraphDatabase.driver(NEO4J_URI,
    auth=(NEO4J_USERNAME, NEO4J_PASSWORD))

llm = AzureOpenAILLM(model_name="gpt-4o-2", ...)

embedder = AzureOpenAIEmbeddings(model="
   "text-embedding-3-small",
```

4 Upload PDF Documents

Upload one or more PDF files

Drag & drop or click to upload

Example files

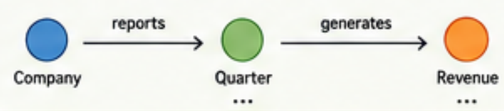
- 10-K_Alphabet_Q1-2023.pdf
- 10-K_Tesla_Q1-2023.pdf
- ...

5 Build Knowledge Graph from PDFs

PDFs → Text Splitter → LLM Extraction → Graph (Nodes & Relationships)

SimpleKGPipeline

- **Entities:** Company, Quarter, Revenue, Net Income, ...
- **Relations:** reports, generates, is part of, affects, ...
- **Prompt:** Extract entities & relationships as JSON
- **Chunking:** FixedSizeSplitter (500 / 100 overlap)



✓ Nodes and relationships stored in Neo4j

6 Create Vector Index & Ask Questions (GraphRAG)

Vector Index

Label: Chunk
Property: embedding
Dimensions: 1536
Similarity: cosine

```
create_vector_index(
  driver, "vector-index-name",
  label="Chunk",
  embedding_property="embedding",
  dimensions=1536,
  similarity_fn="cosine"
)
```

Ask a question

Compare Q1-2023 revenues between Alphabet Inc and Tesla



GraphRAG Pipeline

Retriever (top_k=3) → Context (Chunks + Graph) → LLM (Answer)

Answer (Example)

Company	Quarter	Revenue (USD)	Source
Alphabet Inc	Q1-2023	$69.79B	10-K_Alphabet_Q1-2023.pdf
Tesla, Inc.	Q1-2023	$23.33B	10-K_Tesla_Q1-2023.pdf

Insight: Alphabet's revenue in Q1-2023 was ~3x higher than Tesla's.

7 Visualize Your Graph in Neo4j

Explore nodes and relationships interactively!

Company
Quarter
Revenue
Net Income
...



✓ Use Cypher queries in Neo4j Browser or yfiles_jupyter_graphs_for_neo4j

8 Backup, Download & Restore Neo4j Database

Dump DB

```
!nj/bin/neo4j-admin
 database dump neo4j
```

Download

```
from google.colab
 import files
files.download(
  'neo4j.dump')
```

File Size

```
!du -sh neo4j.dump
```

120M neo4j.dump

Restore DB

```
!nj/bin/neo4j-admin
 database load neo4j
 --from-path=/content/
 --overwrite-dest=true
```

9 Useful Colab Integrations

Forms & Inputs | Camera Capture | JS ↔ Python Communication | BroadcastChannel Cross-output | Google Drive | Google Sheets → Pandas | File Uploads & Downloads | BigQuery Analytics

10 External Tools & Libraries

- matplottib-venn
- cartopy (maps)
- graphviz & pydot
- libarchive (7zip)
- flask, flask-ngrok (Web APIs)
- gspread, gsutil, PyDrive
- yfiles_jupyter_graphs_for_neo4j
- neo4j-graphrag, rapidfuzz

11 Web API with Flask + NGROK

Flask App → NGROK Public URL → REST API (GET, POST, PUT, DELETE)

12 Key Benefits

- ✓ Build rich knowledge graphs from your documents
- ✓ Combine LLM + Graph for smarter answers
- ✓ Vector search + Graph context = powerful RAG
- ✓ Visualize, query, backup, and deploy easily
- ✓ All in Google Colab!



Pro Tip

Use structured graphs + LLMs to turn unstructured PDFs into actionable insights!

